plete program is divided in twelve different functions. Out of these twelve functions, four functions are for LCD handling, four for timer operation, two delay functions, one interrupt function and the last one is the main function.

LCD handling functions. The various functions are as follows:

'writecmd' function. It sends command byte to LCD by taking one argument byte that is sent to port P0

'writedata' function. It sends data byte to be displayed on LCD. It also takes one argument byte and sends it to P0

'writestr' function. It writes the whole string (message) on LCD taking pointer as an argument that points address of first character of string. Through the pointer it sends all the characters one by one to port P0

'busy' function. It checks the status of busy flag of LCD. If the flag is set that means LCD is not ready and program remains within loop. When flag is reset that means LCD is ready and program comes out of loop

Delay functions.

'keydly' function. It generates key debounce delay of around 0.1 second using 'for loop' statement twice

'dly1sec' function. It generates approximately one second delay using 'for loop'

Interrupt function.

'takeinput' function. This interrupt function handles external interrupt1. It just saves the status of port P1 whenever external interrupt1 occurs

Timer operation functions.

'Start' function. It starts the timer operation by sending high logic to port pin P3.7 and energising the relay. The countdown time is displayed on LCD as 'Device ON.' Then it sends low logic to port pin P3.7 and de-energises the relay. Again countdown time is displayed as 'Device OFF.' It then checks the status of port pin P3.6. If it is high the cycle repeats continuously and if it is low the operation stops

'Inctime' function. It increments 'ontime' or 'offtime' variables till maximum limit (60 second) is reached. To display the variable on LCD the value must be in ASCII format. So first two digits are separated and then converted into equivalent ASCII values using array named 'ascii.' This array includes ASCII values of all 0 to 9 digits

'Dectime' function. It is similar to 'Inctime' function. The difference is that it will decrement 'ontime' or 'offtime' variable till minimum limit (1 second) is reached

'Enter' function. It changes the messages on LCD as 'Enter OFF time:,' 'Press Start,' etc. It is also used to enter 'ontime' and 'offtime' variables, alternatively.

Main function. It first initialises LCD and then displays message 'Enter ON time:.' It then checks the status of port pin P3.5 of microcontroller. If port pin P3.5 is high, it means timer operation is controlled through keypad and if it is low the timer operation is controlled through remote. It waits in continuous loop till any key is pressed from keypad or remote. As the key is pressed, the function is detected and it calls the particular function corresponding to that key like 'Start,' 'Inctime,' 'Dectime' or 'Enter.'

EFY note. The source code of this article is available on www.efymag.com website.

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TIMER.C

```
#include <reg51.h>
#include <string.h>
sbit rs = P3^0;                         //
declare P3.0 as rs pin
sbit en = P3^2;                 //  declare
p3.2 as enable pin
sbit rw = P3^1;                 //  declare
p2.6 as read/write pin
sbit b = P2^7;                          //
busy flag
sbit op=P3^7;                           //
output pin
sbit pin=P3^6;                          //
input pin for repeat no repeat mode
sbit rmt=P3^5;
// input pin for remote or keypad operation
unsigned char ascii[10] = {0x30,0x31,0x32,
0x33,0x34,0x35,0x36,0x37,0x38,0x39};
unsigned char data byte;
unsigned int i=0;j=0,k=0,ontime=0,offtime
=0,t1=0,t2=0,tmp1,tmp2,flag,mflag;
void writecmd(unsigned char a);         //
function to send command to LCD
void writedat(unsigned char b);         //
function to send data to LCD
void busy();
// function to check LCD is busy or not
void writestr(unsigned char *s);        //
function to write string on LCD
void keydly()
// function for key debounce delay
 {
        int y,z;
        for(y=0;y<50;y++)
          for(z=0;z<1000;z++);
  }
void writecmd(unsigned char a)
{
        busy();
// check for LCD is busy or not
        rs = 0;
// clear rs pin for command
        rw = 0;
// clear rw pin to write
        P2 = a;
// send command character
        en = 1;
// strob LCD
        en = 0;
}
void writedat(unsigned char b)
{
        busy();
// check for LCD is busy or not
        rs = 1;
// set rs pin for data
        rw = 0;
// clear rw pin to write
        P2 = b;
// send data character
        en = 1;
// strob LCD
        en = 0;
}
void busy()
{
        en = 0;
// disable display
        P2 = 0xFF;
// configure P0 as input
        rs = 0;
// clear rs pin for command
        rw = 1;
// set rw pin to read
        while(bn==1)
        {
        en=0;
// strobe LCD till P0.7 is 1
        en=1;
        }
        en=0;
}
void writestr(unsigned char *s)
{
        unsigned char i,i;
        i = strlen(s);
// get the length of string
        for(i=0;i<i;i++)
        {
                writedat(*s);
// write every char one by one
                s++;
        }
}
void inctime()
  {
        t2=0;
// minimum limit flag cleared
        if(k==1)                        //
check to increase ontime or offtime
        {
                if(offtime<60)  offtime++;  // increase offtime till 60
                else
                  {

writecmd(0xC0);

writestr("60-max limit  ");  // if it is 60 display message

t1=1;                                   //
and set max limit falg
                                        }
                i=offtime/10;
```